Baseline vs. Cross-Encoder Re-Ranking in RAG

RAG Assignment Report

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1 Objective and experimental setup

The goal is to compare two retrieval pipelines that share one chunked document and one LLM. The baseline passes the bi-encoder's Top-3 chunks to the LLM. The second pipeline re-ranks the same Top-10 with a Cross-Encoder and passes its Top-3. The comparison looks for queries where the re-ranking measurably restores relevance.

The document was chosen because it is a dense rulebook: general rules followed by exceptions, prohibitions ("you can't claim…"), role restrictions, footnotes inside tables, and near-duplicate passages (the kidnapped-child rule appears three times for different filing statuses). These are the conditions under which a bi-encoder's single-vector similarity tends to rank the wrong passage first.

Item	Choice
Document	IRS Publication 501 (2025), *Dependents, Standard Deduction, and Filing Information*, 31 pages, `https://www.irs.gov/pub/irs-pdf/p501.pdf` (PDF created 4 Feb 2026; U.S. government work, public domain). SHA-256 `288b62a4...8ff6`
Extraction	PyMuPDF with de-hyphenation; footer, page numbers and margin icons removed; page number kept per word
Chunking	150-word sliding window, 30-word overlap, across page breaks → **292 chunks** (36 span two pages)
Bi-encoder	`sentence-transformers/all-MiniLM-L6-v2`, normalised embeddings, FAISS `IndexFlatIP` (= cosine)
Cross-Encoder	`cross-encoder/ms-marco-MiniLM-L-6-v2`
LLM	Google **Gemini 3.5 Flash** (`gemini-3.5-flash`, free API tier): temperature 0, seed 42, max 256 output tokens, thinking disabled (with it on, hidden reasoning used up the 256-token budget and answers were cut off); same system prompt and template for both pipelines. An earlier full run with local Qwen2.5-1.5B-Instruct is kept for comparison
Hardware	Linux x86-64 laptop CPU, 12 PyTorch threads, no GPU; Python 3.10, torch 2.7.0, sentence-transformers 5.1.2

The chunk size keeps every chunk under the bi-encoder's 256-token limit (median 192 tokens, max 255, none truncated). The free Gemini tier allows 20 requests per model per day, so answers were generated for the 8 key queries only and cached. Retrieval and re-ranking were evaluated on all 16.

2 Architecture comparison

A **bi-encoder** encodes the query and each chunk *separately* into fixed vectors and compares them with cosine similarity. Chunk vectors are computed once, offline, so a query costs one encoder pass plus a vector search. The weakness is that each chunk is compressed into one 384-d vector *before* the query is known. Fine details — a "can't", a footnote, or one sentence inside a passage about something else — get averaged away, and surface word overlap counts for a lot.

A **Cross-Encoder** reads the query and a chunk *together* in one transformer pass (`[CLS] query [SEP] chunk [SEP]`), so self-attention connects every query token to every chunk token before it outputs one relevance score. This is much better at judging whether the passage actually answers the question. However, nothing can be pre-computed, so each (query, chunk) pair costs a full forward pass at query time.

	Pipeline 1 — Baseline	**Pipeline 2 — Re-ranking**
1	embed query, FAISS Top-10	the **same** Top-10 list
2	keep ranks 1–3	Cross-Encoder scores all 10 pairs, sort descending, keep Top-3
3	LLM → Answer 1	same LLM, prompt and settings → Answer 2

The code calls the bi-encoder once per query and builds both contexts from that one list. Original ranks are stored next to the new ones. Unit tests check this (30 tests, all passing).

3 Methodology

- **Queries:** 16, written after reading the PDF for exceptions, prohibitions, role limits, table footnotes, numeric limits and IRS wording that differs from everyday words. All ten retrieved passages were inspected before and after re-ranking; queries with no change were kept as controls.
- **Relevance:** each query's ground truth is a sentence copied from the PDF with its page; a chunk is relevant if it contains it. A query is *improved* if that chunk enters the Top-3 only after re-ranking, *regressed* if the reverse.
- **Answers:** generated only for improved and regressed queries (where the two pipelines give the LLM different evidence), and judged against the chunks each pipeline supplied.
- **Timing:** `time.perf_counter()`, models pre-loaded, 3 warm-ups, then 20 runs × 16 queries per stage; generation from the 16 real API calls, network included, rate-limit waits excluded.
- **Answer cache:** answers are stored in `results/llm_cache.json` under a hash of model, prompts and settings, so a re-run reuses them without spending API quota.

4 Failure mode analysis

Each subsection quotes a requirement from Section 3 of the assignment and gives the query that satisfies it. Ranks and scores are from `results/candidate_results.json`; answers are from Gemini 3.5 Flash with the identical prompt.

Over all 16 queries, the answer sentence was in the Top-3 for **9/16** queries before re-ranking and **15/16** after (Hit@3); MRR@10 rose from **0.574** to **0.849**. The queries were chosen to find failures, so these figures describe this stress-test set, not typical questions.

4.1 At least two failure queries where re-ranking restores relevance

Satisfied by Q16, Q01, Q02, Q03 and Q05. In each, the answer chunk was at bi-encoder rank 4–6, all three baseline chunks were off-topic so Gemini replied "the provided context is insufficient", and the Cross-Encoder moved the answer chunk to **rank 1**, after which Gemini answered correctly with the right page. Q02 and Q03 are two wordings of one question, so they count as one failure tested twice.

4.2 Keyword Trap / Distractor

> *"A query with high lexical overlap to an irrelevant passage, while the true answer uses distinct terminology or synonyms."*

Overview — which query satisfies which requirement

Assignment requirement	Query	Relevant chunk, page	Rank before → after	Answer 1 → Answer 2
4.1 At least two failure queries	Q16, Q01, Q02/Q03, Q05	see below	4–6 → **1**	insufficient → correct
4.2 Keyword Trap / Distractor	**Q16** (synonym), Q01 (distractor)	C095, p.11	4 → **1**	insufficient → correct
4.3 Exceptions & Corner Cases	**Q02/Q03** (penalty), Q04 (exception)	C014, p.3; C100, p.11	6 → **1**; 5 → **1**	insufficient → correct; vague → precise
4.4 Negation & Constraints	**Q05** (role limit)	C089, p.10	5 → **1**	insufficient → correct
4.5 Other Document-Specific	**Q06** (chunk split), plus all seven above	C096, p.11	8 → **3**	correct → correct
4.6 Counter-example	Q08	C060, p.8	**1** → **4**	correct → insufficient

Q16: "Can I claim my live-in nanny who works for me as a dependent?" "Nanny" never appears in the PDF; the rule (C095, p.11) says *"Housekeepers, maids, or servants. If these people work for you, you can't claim them as dependents."* Meanwhile "live-in" matched passages about who lives with you, and the baseline Top-3 was all such distractors: C068 (spouse/child in your home, p.8), C182 (member-of-household test, p.19), C088 (Table 4 rows on relatives living with you, p.10). C095 moved **#4 → #1** (cosine 0.547; CE 2.97 vs. 0.38, −0.30, −1.41 for the distractors).

> **A1:** "... the passages do not contain information regarding whether a live-in nanny... can be claimed as a dependent... the provided context is insufficient."
>
> **A2:** "... no, you cannot claim a housekeeper, maid, or servant who works for you as a dependent (p. 11)."

Verdict: restored; A2 is correct and cited, though it does not say explicitly that a nanny counts as a household employee. **Supporting case, Q01** ("Is a housekeeper who lives in my home my dependent?"): same distractor (head-of-household text C074, C068, C065) but no synonym gap; C095 #4 → #1; A1 "insufficient", A2 "no... you cannot claim them as dependents (p. 11)".

4.3 Exceptions & Corner Cases

> *"The baseline retrieves a general rule chunk, but misses a specific restriction, penalty, or 'except when...' condition located elsewhere."*

Q02/Q03: "What happens if I willfully fail to file a tax return?" / "What is the penalty for failing to file a required return?" The baseline Top-3 held the general filing rules: Table 3 "You must file a return if..." (C032), "Who Should File" (C030), the filing-requirement table (C013). The penalty sits in C014 (p.3), a chunk mostly about defining gross income: *"You may have to pay a penalty if you are required to file a return but fail to do so. If you willfully fail to file a return, you may be subject to criminal prosecution."* C014 moved **#6 → #1** for both wordings (cosine 0.481/0.393; CE 5.56/4.11).

> **Q02 A1:** "... there is no information provided about what happens if you willfully fail to file... The provided context is insufficient."
>
> **Q02 A2:** "If you willfully fail to file a tax return, you may be subject to criminal prosecution (p. 3)."

Verdict: restored for both wordings. Q03's A1 was also "insufficient"; its A2 gave both the penalty and the prosecution sentence (p. 3). The PDF gives no penalty amount and neither answer invented

one. **Supporting case, Q04** ("Does a person who is not a U.S. citizen qualify as a dependent?"): the baseline had the general rule (Table 5, `C104`, p.12) but not the "Exception for adopted child" text (`C100`, p.11), which moved #5 → #1. A1 only said an exception exists (from a cross-reference in `C181`); A2 gave its conditions (adopter is a U.S. citizen or national; child legally adopted or placed and lived with you all year) — a modest gain.

4.4 Negation & Constraints

> *"Queries with qualifiers such as 'not allowed', 'prohibited', or specific role limits where Bi-Encoder embeddings blur negative and positive context."*

Q05: "Is a person I can claim only under a multiple support agreement a qualifying person for head of household?" The answer is Table 4, footnote 5 (`C089`, p.10): *"...that person **isn't** a qualifying person."* The bi-encoder ranked first `C213` (cosine 0.691), which says the **positive**: "You **can** claim someone as a dependent under a multiple support agreement..."; the negative footnote came fifth (0.641). The embedding matched the topic but ignored "can" vs. "isn't". `C089` moved **#5 → #1** (CE 7.23, highest of the ten).

> **A1:** "...the passages do not state whether a person claimed... only under a multiple support agreement qualifies... the provided context is insufficient."
>
> **A2:** "No. A person that you can claim as a dependent only because of a multiple support agreement is **not** a qualifying person for head of household filing status (pp. 10-11)."

Verdict: restored. With the same baseline chunks, local Qwen2.5-1.5B answered "**Yes**" — the opposite of the rule. Q16 and Q01 are also negations ("you can't claim them").

4.5 Other Document-Specific Failures

> *"Any scenario where a vital chunk is buried at ranks 4–10 initially, but gets successfully promoted into the Top 3 by the Cross-Encoder."*

All seven improved queries fit; the table shows what about this document buried each chunk.

Query	Vital chunk	Rank	Why it was buried
Q06	`C096`, p.11	**8 → 3**	Rule and exception split across a chunk boundary (`C096`/`C097`)
Q16, Q01	`C095`, p.11	4 → 1	One-line rule inside a chunk mostly about the child tax credit
Q02/Q03	`C014`, p.3	6 → 1	Two sentences inside a chunk mostly defining gross income
Q05	`C089`, p.10	5 → 1	A table footnote, not body text
Q04	`C100`, p.11	5 → 1	Exception in a different section from the summary table

Q06 ("Can a dependent claim another person as a dependent?") is the purely document-specific case: the rule *"If you can be claimed as a dependent by another taxpayer, you can't claim anyone else as a dependent"* ends `C096`, and its refund-only exception starts `C097`. The baseline had only `C097`; re-ranking brought `C096` in from rank 8. Both answers were correct — A1 inferred the rule from the exception, A2 quotes it (p. 11) — so the gain is in grounding, not correctness.

4.6 Counter-example (Q08) and controls

For "What is the capital loss deduction limit for married filing separately?" the baseline had *"Your capital loss deduction limit is $1,500 (instead of $3,000 on a joint return)"* (`C060`, p.8) at rank 1 and A1 was correct. The Cross-Encoder preferred broader married-filing-separately chunks (`C062` 3.63, `C063` 3.42, `C252` 2.68 vs. `C060` 2.61), `C060` fell to **#4**, and A2 said the context was insufficient

(Qwen invented "$15,750"). In the 8 controls (Q07, Q09–Q15) the answer chunk was in both Top-3s; re-ranking sometimes moved it within the Top-3 (Q07: `C249` 9 → 1) but did not change the evidence, and in the Qwen run both pipelines answered them alike.

5 Latency profiling

Stage (median)	**Baseline**	**Re-ranking**
Stage 1: embed query + FAISS Top-10	8.9 ms	8.9 ms
Stage 2: Cross-Encoder on 10 pairs	—	260.2 ms
Retrieval total	**8.9 ms**	**269.1 ms**
LLM generation (Gemini API, incl. network)	3.20 s	2.43 s
End-to-end	3.21 s	2.74 s

Retrieval figures: $n = 320$ each (stdev 2.3 ms and 25.4 ms). Generation figures: $n = 8$ per pipeline.

Re-ranking makes retrieval about **30× slower** (≈26 ms per pair, a 6-layer transformer pass on up to 512 tokens). That is about 10% of the re-ranking pipeline's end-to-end time with Gemini. With the local CPU model (∼34 s per answer) it was under 1%, so the faster the generator, the more the reranker's cost matters.

CPU timings vary with machine load. Across five runs on this laptop, the reranker measured 171–264 ms and Stage 1 7.9–13.1 ms (ratio 20–31×). Generation time depends on the API and on answer length, not on the pipeline. One 36 s API outlier is why medians are reported.

6 Scaling question: why not Cross-Encode the whole corpus?

A Cross-Encoder's score depends on the query and the chunk together, so nothing can be indexed ahead of time. Every query would need one transformer pass **per chunk in the collection**. A bi-encoder pays the per-chunk cost once, offline, and answers queries with one embedding plus a vector search, which approximate indexes keep fast even for millions of vectors.

Measured on this corpus: Cross-Encoding all 292 chunks for one query took **9.0 s** (≈31 ms per pair). The bi-encoder search took 9 ms, plus 260 ms for re-ranking 10 candidates. At the same rate, 10,000 chunks would take about 5 minutes per query and 1 million about 8.6 hours. For Q01 the full-corpus pass chose the same best chunk (`C095`) as the two-stage pipeline.

Retrieving a small candidate set first keeps the expensive model's cost fixed (k pairs), whatever the corpus size. The price is that anything outside the bi-encoder's Top-k cannot be recovered, which is why practical systems re-rank a larger pool (e.g. 50–100).